

Name: _____

Teacher/Class: _____

SYDNEY TECHNICAL HIGH SCHOOL

YEAR 12

HSC ASSESSMENT TASK 3

2005

EXTENSION 1 MATHEMATICS

Time Allowed: 70 minutes

Instructions:

- Write your name and class at the top of each page.
- All necessary working must be shown. Marks may be deducted for careless or badly arranged work.
- Marks indicated are a guide only and may be varied if necessary.
- Start each question on a new page.

Question 1	Question 2	Question 3	Question 4	Question 5	Question 6	Total
/10	/10	/11	/10	/11	/10	/62

Question 1 (10 marks)

a) Differentiate

(i) xe^{x^2}

(ii) $\log_{10} x$ (4)

b) Find

(i) $\int 4e^{2x+1} dx$

(ii) $\int \tan^2 \theta d\theta$ (3)

c) Show that $\frac{d}{dx}(\sin^3 x) = 3 \cos x - 3 \cos^3 x$ and hence, or otherwise find

$$\int (\cos x - \cos^3 x) dx$$
 (3)

Question 2 (10 marks)

a) Find the exact value of (3)

(i) $\log_2 \sqrt[3]{4}$

(ii) $\sin^{-1}\left(-\frac{1}{\sqrt{2}}\right)$

b) State the domain and range of $y = 4 \sin^{-1}(1 - x)$ (2)c) Find $\int \sin^2 4x dx$ (2)d) Find $\int \frac{4}{\sqrt{9 - 4x^2}} dx$ (3)

Question 3 (11 marks)

a) Differentiate (4)

(i) $\tan^{-1}\left(\frac{x}{2}\right)$

(ii) $\sin^{-1}(x^2)$

b) Solve $\log_8 2 = \log_x 5$ (2)

c) (i) Express $\sqrt{3} \sin x - \cos x$ in the form of $R \sin(x - \alpha)$, where (5)

$R > 0$ and $0 < \alpha < \frac{\pi}{2}$

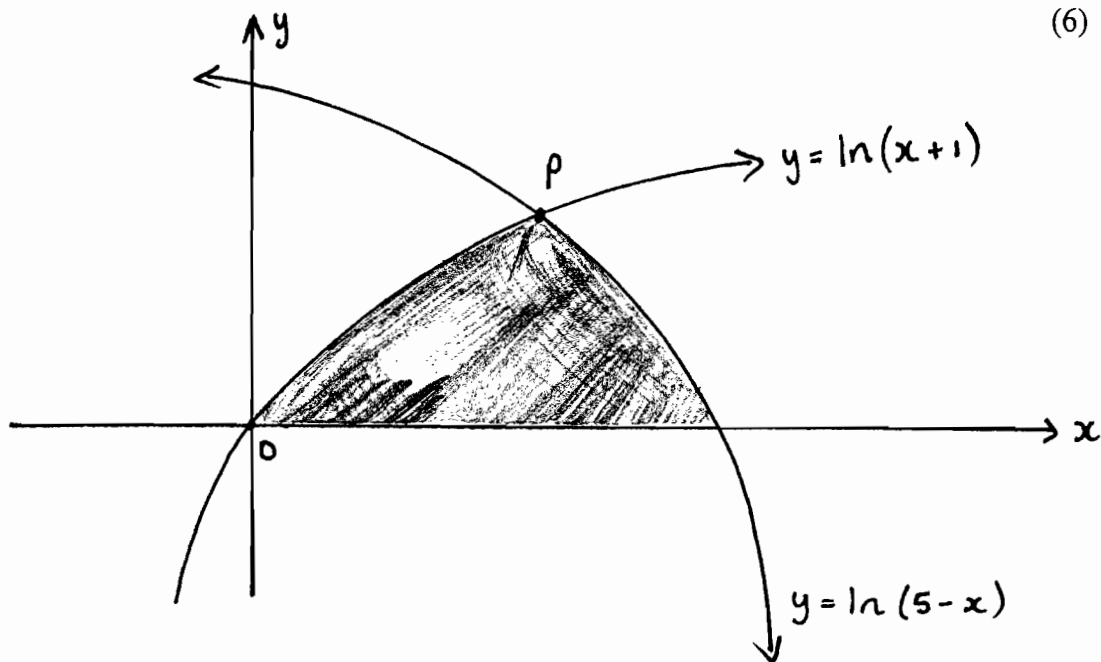
(ii) Sketch the graph of $y = \sqrt{3} \sin x - \cos x$ for $0 \leq x \leq 2\pi$, showing intercepts and endpoints.

(iii) Use your sketch to state the **number** of solutions to the equation $\sqrt{3} \sin x - \cos x = -1$.

Question 4 (10 marks)

a)

(6)



The diagram shows the graphs of $y = \ln(x+1)$ and $y = \ln(5-x)$ intersecting at **P**.

- (i) Show that the coordinates of **P** are $(2, \ln 3)$
- (ii) The area, **A**, enclosed by the curve $y = \ln(x+1)$, the **y axis** and the lines $y = 0$ and $y = \ln 3$ is given by the integral,

$$A = \int_0^{\ln 3} (e^y - 1) dy$$

Use this integral to calculate the exact area, **A**.

- (iii) Calculate the exact area of the shaded region shown above.

b) (i) Sketch $y = \cos^{-1} x$ showing all intercepts and end points (3)

- (ii) Hence find $\int_{-1}^1 \cos^{-1} x dx$

c) Write down the general solution for $\tan x = \frac{1}{\sqrt{3}}$ (1)

Question 5 (11 marks)

a) Given $\int_0^{\frac{\pi}{4}} \cos^2 \theta \, d\theta = \frac{\pi+2}{8}$ (3)

use the substitution $x = 2 \sin \theta$, to evaluate $\int_0^{\sqrt{2}} \sqrt{4-x^2} \, dx$

b) $y = \sin x + \sin x \cos x$ for $0 \leq x \leq 2\pi$ (8)

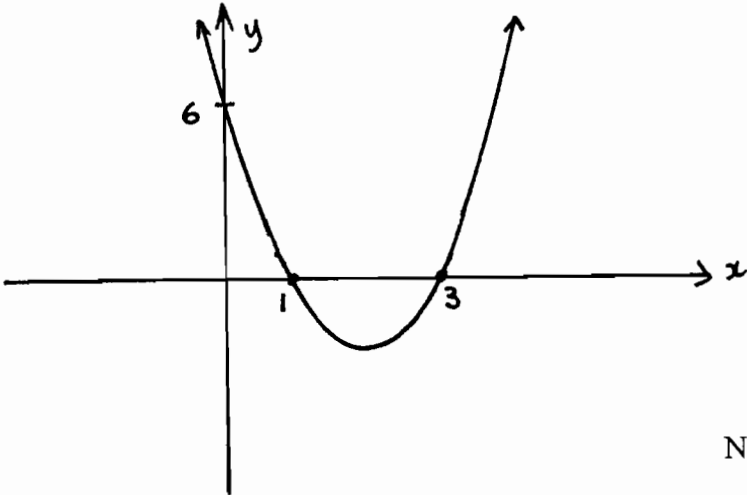
(i) Show that $\frac{dy}{dx} = \cos x + \cos 2x$

(ii) Find the co ordinates of the three stationary points and determine their nature.

(iii) Sketch the curve for the given domain.

Question 6 (10 marks)

a) Find $\int \frac{x+1}{x^2+4} \, dx$ (3)

b)  (7)

Not to scale

(i) Show that the equation of the parabola $y = f(x)$ is given by

$y = 2x^2 - 8x + 6$ and state the co ordinates of the vertex

(ii) State the largest possible domain, that includes $x = 0$, for which $y = f(x)$ has an inverse.

(iii) State the domain of $y = f^{-1}(x)$.

(iv) What is the equation of the inverse function $y = f^{-1}(x)$.

STANDARD INTEGRALS

$$\int x^n dx = \frac{1}{n+1} x^{n+1}, \quad n \neq -1; \quad x \neq 0, \text{ if } n < 0$$

$$\int \frac{1}{x} dx = \ln x, \quad x > 0$$

$$\int e^{ax} dx = \frac{1}{a} e^{ax}, \quad a \neq 0$$

$$\int \cos ax dx = \frac{1}{a} \sin ax, \quad a \neq 0$$

$$\int \sin ax dx = -\frac{1}{a} \cos ax, \quad a \neq 0$$

$$\int \sec^2 ax dx = \frac{1}{a} \tan ax, \quad a \neq 0$$

$$\int \sec ax \tan ax dx = \frac{1}{a} \sec ax, \quad a \neq 0$$

$$\int \frac{1}{a^2 + x^2} dx = \frac{1}{a} \tan^{-1} \frac{x}{a}, \quad a \neq 0$$

$$\int \frac{1}{\sqrt{a^2 - x^2}} dx = \sin^{-1} \frac{x}{a}, \quad a > 0, \quad -a < x < a$$

$$\int \frac{1}{\sqrt{x^2 - a^2}} dx = \ln \left(x + \sqrt{x^2 - a^2} \right), \quad x > a > 0$$

$$\int \frac{1}{\sqrt{x^2 + a^2}} dx = \ln \left(x + \sqrt{x^2 + a^2} \right)$$

NOTE : $\ln x = \log_e x, \quad x > 0$

Teacher's Name: _____

Student's Name/Nº: _____

Question 1

a) i. $y = x e^{x^2}$

$$\frac{dy}{dx} = v u' + u v'$$

$$= e^{x^2} \cdot 1 + x \cdot 2x e^{x^2} \quad \checkmark$$

$$= e^{x^2} (1 + 2x^2)$$

ii. $\log_{10} x = \frac{\ln x}{\ln 10} \quad \checkmark$

$$\therefore \frac{d}{dx} (\log_{10} x) = \frac{1}{x \ln 10} \quad \checkmark$$

b) i. $\int 4 e^{2x+1} dx$
$$= 2 e^{2x+1} + C \quad \checkmark$$

ii. $\int \tan^2 \theta d\theta$
$$= \int (\sec^2 \theta - 1) d\theta \quad \checkmark$$

$$= \tan \theta - \theta + C \quad \checkmark$$

c) $\frac{d}{dx} (\sin^3 x)$
$$= 3 (\sin x)^2 \cdot \cos x \quad \checkmark$$

$$= 3 \cos x (1 - \cos^2 x)$$

$$= 3 \cos x - 3 \cos^3 x \quad \checkmark$$

$$\therefore \int \cos x - \cos^3 x dx \quad \checkmark$$

$$= \frac{1}{3} \sin^3 x + C$$

Question 2

a) i. $\log_2 \sqrt[3]{4} = \log_2 (2^2)^{1/3}$
$$= \frac{2}{3} \log_2 2$$

$$= \frac{2}{3} \quad \checkmark$$

ii. $\sin^{-1}(-1/\sqrt{2}) = -\pi/4 \quad \checkmark$
① = sign

b) $y = 4 \sin^{-1}(1-x)$

D: $-1 \leq 1-x \leq 1$

$$-2 \leq -x \leq 0$$

$$2 \geq x \geq 0$$

$$\therefore 0 \leq x \leq 2 \quad \checkmark$$

R: $-2\pi \leq y \leq 2\pi \quad \checkmark$

c) $\int \sin^2 4x dx$

$$= \frac{1}{2} \int 1 - \cos 8x dx \quad \checkmark$$

$$= \frac{1}{2} \left[x - \frac{1}{8} \sin 8x \right] + C \quad \checkmark$$

d) $\int \frac{4}{\sqrt{9-4x^2}} dx$

$$= 4 \int \frac{1}{\sqrt{4(9/4-x^2)}} dx \quad \checkmark$$

$$= 4 \int \frac{1}{2 \sqrt{9/4-x^2}} dx \quad \checkmark$$

$$= 2 \sin^{-1} x / \frac{3}{2}$$

$$= 2 \sin^{-1} \frac{2x}{3} + C \quad \checkmark$$

Teacher's Name:

Student's Name/N^o:

Question 3

$$a) \frac{d}{dx} \left(\tan^{-1} \frac{x}{2} \right) = \frac{2}{x^2 + 4} \quad \checkmark$$

$$ii. \sin^{-1}(x^2) = y \quad \text{let} \\ u = x^2$$

$$y = \sin^{-1} u$$

$$\frac{dy}{dx} = \frac{dy}{du} \cdot \frac{du}{dx}$$

$$= \frac{1}{\sqrt{1-u^2}} \cdot 2x \quad \checkmark$$

$$= \frac{2x}{\sqrt{1-x^4}} \quad \checkmark$$

$$b) \log_8 2 = \log_8 8^{1/3} = 1/3$$

$$\text{or } \log_x 5 = 1/3 \quad \checkmark$$

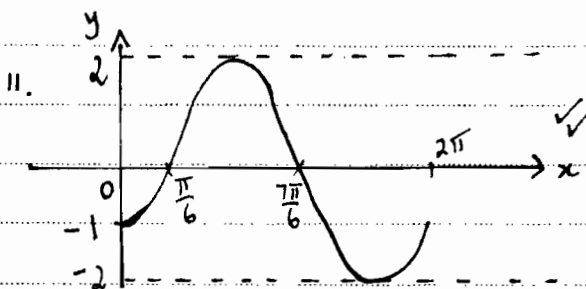
$$x^{1/3} = 5 \\ x = 125 \quad \checkmark$$

$$c) R = \sqrt{3^2 + 1^2} \\ = 2$$

$$\tan \alpha = 1/\sqrt{3}$$

$$\alpha = \pi/6 \quad \checkmark$$

$$\therefore \sqrt{3} \sin x - \cos x = 2 \sin(x - \frac{\pi}{6})$$



iii.

three

Question 4

$$a) i. \ln(x+1) = \ln(5-x)$$

$$x+1 = 5-x \quad \checkmark$$

$$2x = 4$$

$$x = 2 \quad y = \ln 3$$

$$\therefore P(2, \ln 3)$$

$$ii. \int_0^{\ln 3} e^y - 1 \, dy$$

$$= [e^y - y]_0^{\ln 3} \quad \checkmark$$

$$= e^{\ln 3} - \ln 3 - (e^0 - 0)$$

$$= 3 - \ln 3 - 1$$

$$= 2 - \ln 3 \quad \checkmark$$

$$iii. y = \log_e(5-x) \rightarrow e^y = 5-x$$

$$x = 5 - e^y$$

$$A_y = \int_0^{\ln 3} 5 - e^y \, dy \quad \checkmark$$

$$= [5y - e^y]_0^{\ln 3}$$

$$= 5 \ln 3 - 3 - (0 - e^0)$$

$$= 5 \ln 3 - 3 + 1$$

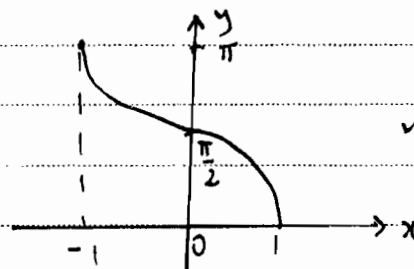
$$= 5 \ln 3 - 2 \quad \checkmark$$

$\therefore$ shaded area

$$= 5 \ln 3 - 2 - (2 - \ln 3)$$

$$= 5 \ln 3 - 2 - 2 + \ln 3$$

$$= 6 \ln 3 - 4 \quad \checkmark$$



$$\cos x \, dx = 1 \times \pi \quad \checkmark$$

Question 4 con +

$$c) \tan x = 1/\sqrt{3}$$

$$x = n\pi + \frac{\pi}{6}$$

or

$$x = n \cdot 180^\circ + 30^\circ$$

where n is any integer $\leftarrow$ must have

✓ ①

Question 5

$$a) \int_0^{\sqrt{2}} \sqrt{4-x^2} dx$$

$$x = 2\sin\theta$$

$$\sqrt{2} = 2\sin\theta$$

$$\theta = \frac{\pi}{4}$$

$$\frac{dx}{d\theta} = 2\cos\theta$$

$$2\sin\theta$$

$$\theta = 0$$

$$\therefore \int_0^{\pi/4} \sqrt{4-4\sin^2\theta} \cdot 2\cos\theta d\theta$$

$$= \int_0^{\pi/4} 2\sqrt{1-\sin^2\theta} \cdot 2\cos\theta d\theta$$

$$= 4 \int_0^{\pi/4} \cos\theta \cdot \cos\theta d\theta$$

$$= 4 \int_0^{\pi/4} \cos^2\theta d\theta$$

$$= 4 \times \left(\frac{\pi+2}{8} \right)$$

$$= \frac{\pi+2}{2}$$

$$b) y = \sin x + \sin x \cos x$$

$$1) y = \sin x + \frac{1}{2} \sin 2x$$

$$\therefore \frac{dy}{dx} = \cos x + \frac{1}{2} \cos 2x \times 2$$

$$= \cos x + \cos 2x$$

✓

$$4) \text{ find } \frac{dy}{dx} = 0$$

$$\cos x + \cos 2x = 0$$

$$\cos x + 2\cos^2 x - 1 = 0 \quad \checkmark$$

$$2\cos^2 x + \cos x - 1 = 0$$

$$(2\cos x - 1)(\cos x + 1) = 0$$

$$\therefore \cos x = \frac{1}{2} \quad \cos x = -1$$

$$\therefore x = \frac{\pi}{3}, \frac{5\pi}{3} \quad x = \pi \quad \checkmark$$

$\left(\frac{\pi}{3}, \frac{3\sqrt{3}}{4}\right)$ Nature

x	1°	$\frac{\pi}{3}$	1.5°
y'	0.12	$-$	-0.9

MAX TP ✓

$\left(\frac{5\pi}{3}, -\frac{3\sqrt{3}}{4}\right)$

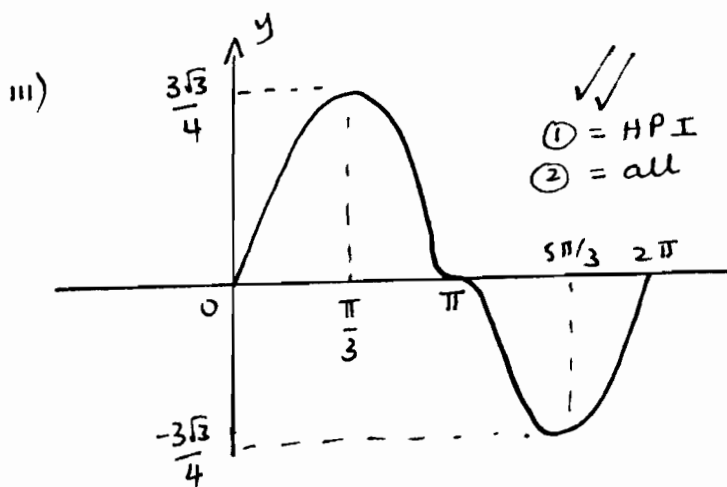
x	5°	$\frac{5\pi}{3}$	5.5°
y'	-0.5	$-$	0.7

MIN TP ✓

$(\pi, 0)$

x	3	π	3.5
y'	-0.02	$-$	-0.1

HPI ✓



Question 6

a) $\int \frac{x+1}{x^2+4} dx$

$$= \int \frac{x}{x^2+4} dx + \int \frac{1}{x^2+4} dx \quad \checkmark$$

$$= \frac{1}{2} \ln(x^2+4) + \frac{1}{2} \tan^{-1} \frac{x}{2} + C \quad \checkmark$$

b) $y = a(x-1)(x-3)$ thru $(0,6)$

i. $6 = a(-1)(-3)$

$$a = 2 \quad \checkmark$$

∴ $y = 2(x-1)(x-3)$

$$y = 2x^2 - 8x + 6$$

vertex $(2, -2) \quad \checkmark$

ii. $x \leq 2 \quad \checkmark$

iii. $y = f^{-1}(x) \quad D: x \geq -2 \quad \checkmark$

iv. $x = 2y^2 - 8y + 6 \quad \checkmark$

$$\frac{x}{2} = y^2 - 4y + 3$$

$$\frac{x}{2} = y^2 - 4y + 4 - 1$$

$$\frac{x}{2} + 1 = (y-2)^2$$

$$y - 2 = \pm \sqrt{\frac{x+2}{2}}$$

$$y = 2 \pm \sqrt{\frac{x+2}{2}} \quad \checkmark$$

$$y = 2 - \sqrt{\frac{x+2}{2}} \quad \checkmark$$

∴ $f^{-1}(x) = 2 - \sqrt{\frac{x+2}{2}}$